

Lab 3, Act 1: Answer Key

Trusting a positive detection

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. Open-ended items (1c, 2c, 3b, and the group discussion) get sample answers or discussion points instead. All numbers are computed in R when this file is knitted.

This Act is discovery-based (no formula is given up front), so expect more variation in *how* students reach a correct answer than in later Acts. Rates used throughout: prevalence = 0.0001, sensitivity = 0.98, false positive rate = 0.01.

Part 1: Count your way to an answer

1a.

Suppose the cameras log $N = 100,000$ trigger events over a season. Fill in the table below using only the rates given above. Every blank cell is just one of those rates applied to a total you already know. Round to the nearest whole number.

Answer.

	Flagged	Not flagged	Total
Actually the species	10	0	10
Not the species	1,000	98,990	99,990
Total	1,010	98,990	100,000

Why. Flagged species events = $10 \times 0.98 = 9.8 \approx 10$. Flagged non-species events = $99,990 \times 0.01 = 999.9 \approx 1,000$. The “not flagged” cells are what’s left in each row.

Watch for. Applying the false positive rate to the 10 species events, or the sensitivity to the 99,990 non-species events. Rows that don’t add up to their totals.

1b.

Of the images the classifier *flagged*, what fraction are actually the target species? No formula needed; just use the counts in your completed table.

Answer. $10/1,010 \approx 0.0099$, about 1%.

Why. The denominator is everything in the “Flagged” column (true and false alarms together); the numerator is the true detections in that column.

Watch for. Dividing by 10 (the species total) instead of by 1,010 (the flagged total), which gives 0.98, the sensitivity. That’s the central confusion of this Act: $P(\text{flagged} \mid \text{species})$ is not $P(\text{species} \mid \text{flagged})$.

1c.

Is this number close to what you expected, given how good the classifier's sensitivity and false-positive rate sounded at the start? In a sentence or two, what would you tell the field team about how to handle a single flagged image?

Sample answer. No, about 1% is far lower than the 98% and 1% rates suggest. A single flagged image is almost always a false alarm, so the team should look for confirmation (a second independent detection, tracks, scat, or a person checking the image) before acting on it.

Why. False alarms come from the huge pool of non-species events; even a 1% false positive rate applied to 99,990 events swamps the 10 real ones.

Watch for. Answers that call the classifier “bad.” Its error rates are fine; the problem is how rare the species is. Saying so here sets up 3b.

Part 2: Turn your counting into a general recipe

2a.

Rewrite the two counts you used in 1b (the number of flagged images that are truly the species, and the total number of flagged images) as expressions using N and the rates given at the top (prevalence, sensitivity, false positive rate), instead of plugged-in numbers.

Answer.

- Flagged and truly the species: $N \times \text{prevalence} \times \text{sensitivity}$.
- All flagged: $N \times \text{prevalence} \times \text{sensitivity} + N \times (1 - \text{prevalence}) \times \text{false positive rate}$.

Why. Each count is “how many events are in that row” times “the chance an event in that row gets flagged,” as in the table.

Watch for. Leaving out the $(1 - \text{prevalence})$ factor, i.e. applying the false positive rate to all N events. With prevalence this small the number barely changes, so ask the student which events can produce a false alarm.

2b.

Divide your two expressions from 2a to reconstruct the fraction from 1b. N should appear in both the top and bottom; cancel it out. What are you left with?

Answer.

$$\text{PPV} = \frac{\text{prevalence} \times \text{sensitivity}}{\text{prevalence} \times \text{sensitivity} + (1 - \text{prevalence}) \times \text{false positive rate}} = 0.0097$$

Why. N multiplies every term, so it cancels. The value differs slightly from 1b (0.0099) only because 1a rounded 9.8 and 999.9 to whole numbers.

Watch for. Students who think the 0.0097 vs. 0.0099 difference means an error. Cancelling a rate instead of N . Stalling while trying to recall “the formula” instead of generalizing their own Part 1 arithmetic.

2c.

State your result from 2b as a general rule, in your own words or symbols: given a classifier's sensitivity, false-positive rate, and the prevalence of the thing it's looking for, how do you compute the chance that a flagged event is real?

Sample answer. “The chance a flag is real equals the true flags divided by all flags, where true flags = prevalence $\times$ sensitivity and false flags = $(1 - \text{prevalence}) \times \text{false positive rate}$.” Any wording or notation that captures the 2b formula is correct.

Why. This is Bayes' rule (the positive predictive value), which students built here before seeing its name in lecture.

Watch for. A rule that restates the 1b numbers instead of **generalizing them.**

Part 3: Try it somewhere else

3a.

A second survey site uses the same classifier, but prevalence is 0.01 (1 in 100 trigger events). Using your recipe from Part 2 (or a fresh counting table), find this second site's PPV.

Answer. $PPV = \frac{0.01 \times 0.98}{0.01 \times 0.98 + 0.99 \times 0.01} = 0.497$, about 50%.

Why. Prevalence is 100 times higher, so true flags (0.0098) are now about as common as false flags (0.0099).

Watch for. Changing the sensitivity or false positive rate too (the handout says they're the same). Students who use a counting table get the same answer (980 true flags vs. 990 false flags out of 100,000).

3b.

Same classifier, same sensitivity, same false-positive rate, but a different answer. What does comparing these two sites tell you about how much those two classifier properties matter versus how common (prevalent) the thing you're looking for actually is? Feel free to use the term "base rate" and speculate on how your findings relate to the "base rate fallacy."

Sample answer. The classifier's properties didn't change, yet the chance a flag is real went from about 1% to about 50%. Prevalence (the base rate) matters as much as the classifier's accuracy. The base rate fallacy is judging a positive result by the test's accuracy alone and ignoring how rare the target is.

Why. When the target is rare, almost every event is a non-target event, so even a small false positive rate produces far more false flags than true ones.

Watch for. Concluding that the classifier "works better" at the second site. Give full credit for any reasonable connection to the base rate fallacy, even if loosely stated.

Group discussion and revision

Compare your Part 2 recipe and your Part 3 answer with your group. Does everyone's recipe agree once you strip away notation differences? If not, find where they diverge. Record any changes, new insights, and muddiest points.

Discussion points. No fixed answer. Look for students who found the sensitivity-vs-PPV confusion in their own or a groupmate's work, or who reconciled different notations for the same recipe.